

Motor Insurance Pricing Model

Frequency-Severity Actuarial Analysis & Commercial Pricing Engine

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Background:	Master's in Finance, Actuarial Science & Data Science
Dataset:	French Motor Third-Party Liability (freMTPL2) — 678,013 policies
Methods:	Negative Binomial GLM · Gamma GLM · XGBoost · Financial Pricing Model

Executive Summary

This report presents an end-to-end actuarial pricing model for motor insurance, built using the industry-standard **Frequency-Severity** methodology. The model predicts expected claim frequency and severity for individual policyholders, combines them into a technically sound Pure Premium, and converts this into a market-ready Commercial Premium after accounting for expenses, profit margin, and reinsurance loading. The analysis was benchmarked against a modern machine learning model (XGBoost) to validate the robustness of the findings.

Metric	Result
Policies analyzed	678,013
Claim rate	5.02% of policies had at least one claim
Best frequency model	Negative Binomial GLM (AIC 285,537 vs Poisson 286,182)
ML benchmark	XGBoost improves accuracy by only ~1.2% — GLM remains preferred for pricing
Strongest risk driver	Bonus-Malus score (prior claims history)
Highest-risk segment	Drivers 18–25 with vehicles 0–2 years old (€258 commercial premium)
Portfolio validation	Pure Premium within 11.5% of actual claims — conservative, well-calibrated

1. Business Problem & Methodology

Insurance companies must price policies accurately: charging enough to cover expected future claims and operating costs, while remaining competitive. This project answers the core actuarial question: *given a policyholder's risk profile, what premium should they be charged?*

The pricing logic follows five stages:

1. **Frequency Model** — predicts the expected number of claims per policy-year (Negative Binomial GLM).
2. **Severity Model** — predicts the average cost per claim, given a claim occurs (Gamma GLM).
3. **Pure Premium** — Frequency $\times$ Severity, the theoretical baseline cost.
4. **ML Benchmark** — XGBoost comparison to test whether a more complex model materially improves accuracy.
5. **Commercial Premium** — Pure Premium adjusted for expenses, profit margin, and reinsurance loading.

1.1 Dataset

The analysis uses the **French Motor Third-Party Liability (freMTPL2)** dataset, a well-known, industry-recognized dataset (CASdatasets) widely used in actuarial research and training. It contains 678,013 policies with driver, vehicle, and regional characteristics, linked to 26,639 individual claim records.

2. Frequency Model Results

A Negative Binomial GLM was selected over a standard Poisson GLM after detecting significant overdispersion in the claim count data (Pearson $\chi^2/df \approx 2.55$). The Negative Binomial model achieved a lower AIC (285,536.93 vs 286,182.38) and lower deviance (189,032 vs 216,840), confirming it better captures the variance structure of claims.

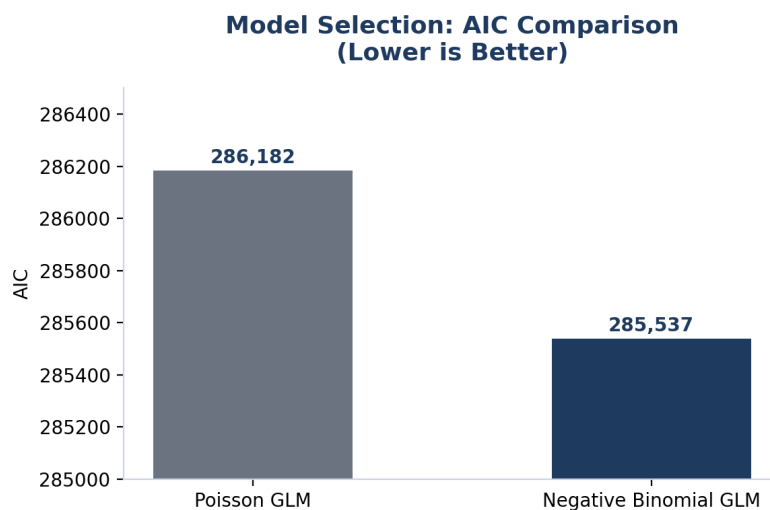


Figure 1 — Model selection based on AIC (lower is better).

The model confirms a well-established actuarial pattern: **young drivers (18–25) show substantially higher claim frequency** than every other age group, at 0.175 claims per policy-year — roughly 80% higher than the

portfolio average.

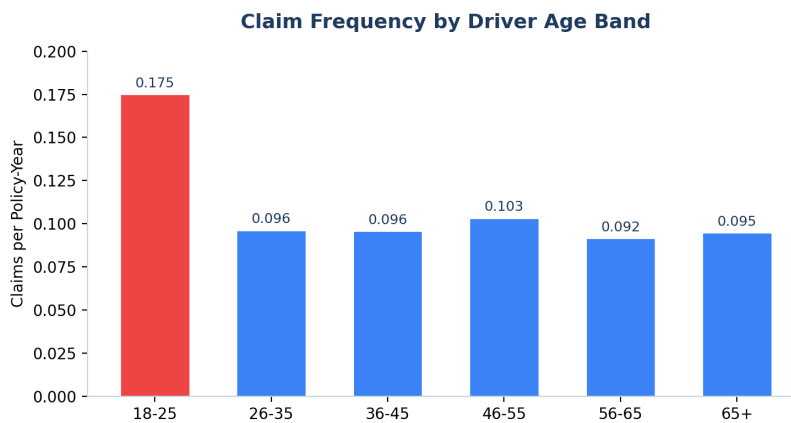


Figure 2 — Observed claim frequency by driver age band.

3. Severity Model Results

Claim severity (the cost of a claim, given it occurs) showed very weak dependence on policyholder features (Pseudo $R^2 = 0.0018$) once large claims were excluded. This is a well-documented finding in actuarial practice: accident cost is driven primarily by random factors — accident circumstances, medical costs, legal outcomes — rather than by static policy attributes. The only significant driver identified was **vehicle age**: newer vehicles cost more to repair.

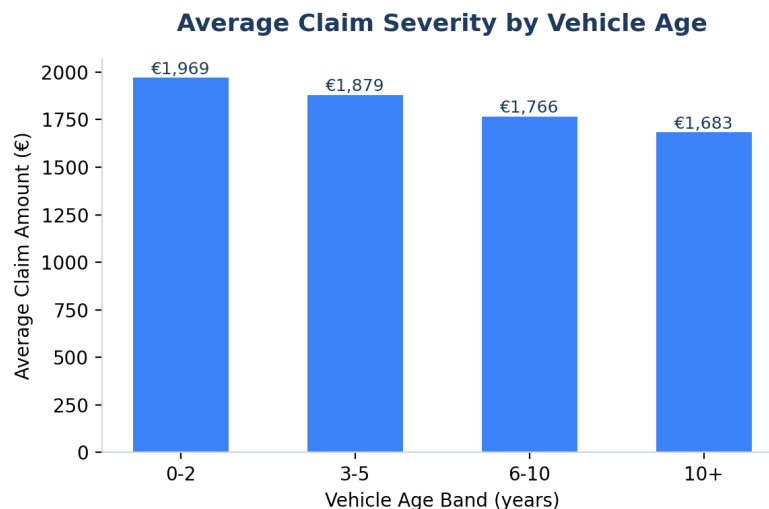


Figure 3 — Average claim severity by vehicle age band.

A small number of extreme claims (42 policies, 0.17% of claims with an amount above €100,000, up to €4.08 million) were separated from the statistical severity model, following standard actuarial practice of handling catastrophic claims via a reinsurance loading rather than individual-level prediction.

4. Machine Learning Benchmark

An XGBoost model was trained on the same frequency prediction task to evaluate the trade-off between interpretability and predictive accuracy — a key consideration in regulated insurance pricing. XGBoost achieved only a marginal improvement in Poisson Deviance (0.3178 vs 0.3216 for the GLM, a ~1.2% gain), reinforcing that the GLM already captures the dominant risk relationships in this dataset.

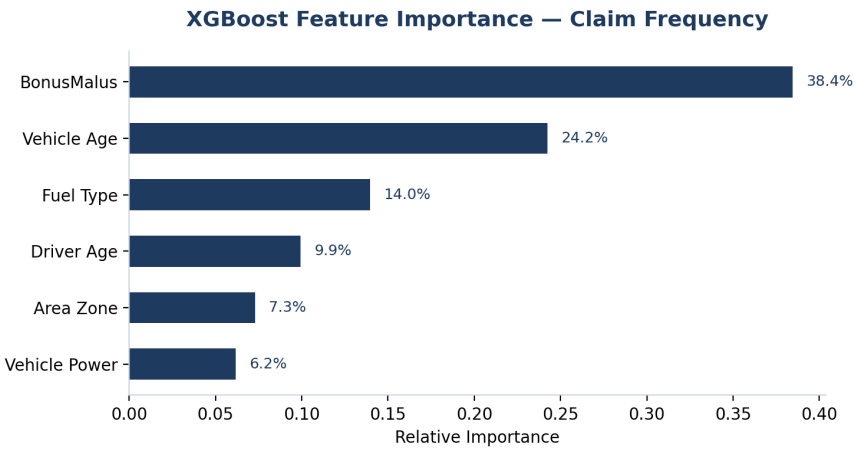


Figure 4 — XGBoost feature importance for claim frequency.

Feature importance rankings from XGBoost closely mirror the statistical significance ordering from the GLM — **Bonus-Malus score** and **vehicle age** dominate in both approaches, cross-validating the key risk drivers identified.

Model	Interpretability	Accuracy (Deviance)	Recommended Use
Negative Binomial GLM	Full — every coefficient explainable	0.3216	Regulatory pricing
XGBoost	Requires SHAP / importance analysis	0.3178	Internal risk scoring, fraud detection

5. Commercial Pricing Results

The Pure Premium was converted into a Commercial Premium using illustrative industry-standard loadings: 20% operating expenses, 10% profit margin, and 5% reinsurance loading — a total markup of 50% over the Pure Premium.

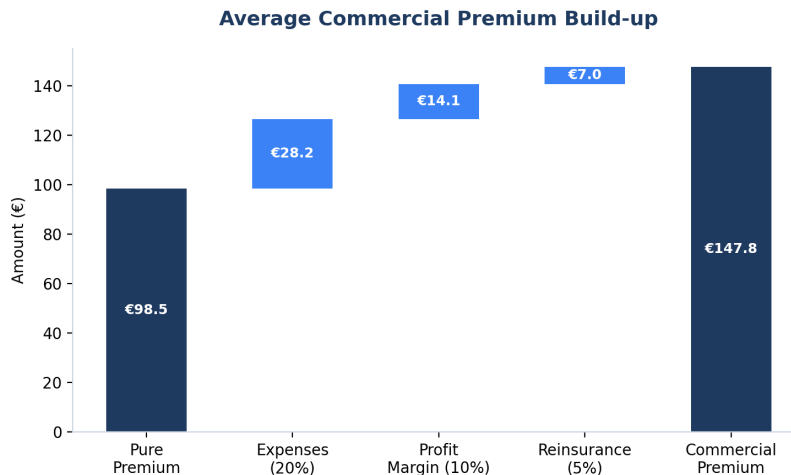


Figure 5 — Average commercial premium build-up (portfolio average).

The resulting pricing engine produces risk-differentiated, commercially sensible premiums. The highest risk segment — young drivers (18–25) with new vehicles (0–2 years) — receives a premium of €258, roughly 57% higher than the lowest-risk segment shown below (€165), driven by both higher claim frequency and higher repair costs for newer vehicles.

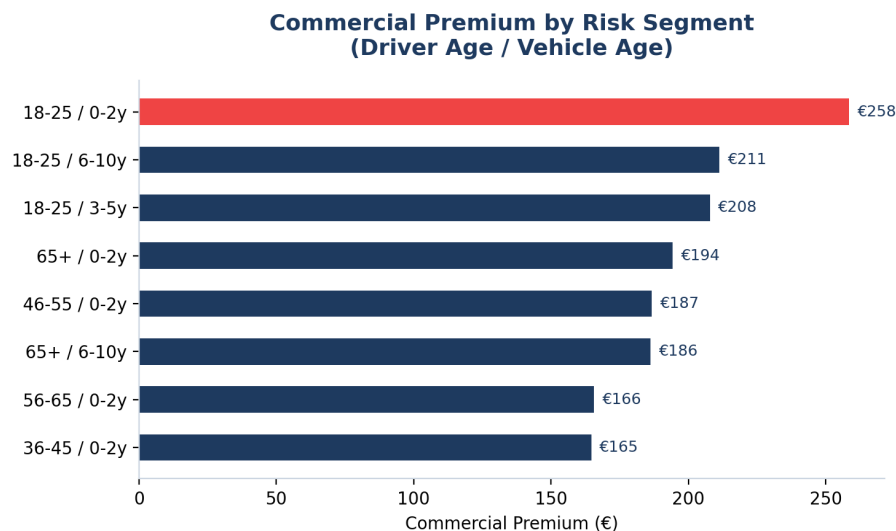


Figure 6 — Commercial premium across key risk segments.

Portfolio validation: total Pure Premium (€66.8M) is 11.5% higher than total actual claims (€59.9M) for the attritional portion of the book. This is expected — actual claims include the 42 large claims excluded from the severity model — and the resulting conservative bias (slightly overpricing rather than underpricing) is a desirable property in insurance pricing.

6. Limitations & Future Work

- **Severity modeling** could be improved with additional variables not present in this dataset (e.g., claim type, injury severity, legal representation).
- **Large claims** were excluded from statistical modeling; a dedicated Extreme Value Theory (EVT) approach could model tail risk more rigorously for reinsurance pricing.
- **Temporal effects** (inflation, seasonality) were not modeled, as the dataset represents a single time period.
- **Interaction effects** (e.g., young driver × high-powered vehicle) were not explicitly tested and could reveal additional risk segments.
- Commercial loading assumptions (20% expenses, 10% profit, 5% reinsurance) are illustrative; production deployment would use company-specific, actuarially-justified figures.

7. Conclusion

This project demonstrates a complete, defensible actuarial pricing pipeline — from raw claims data to a commercially deployable pricing model — combining classical actuarial methodology (GLMs) with modern machine learning benchmarking. The workflow mirrors the actual pricing process used by motor insurers and insurtech companies, and the accompanying interactive dashboard allows real-time premium calculation and risk factor exploration for any policyholder profile.

Full technical implementation (notebook, models, interactive dashboard) available on GitHub:
github.com/abdellahelkhamlichi/motor-insurance-pricing